

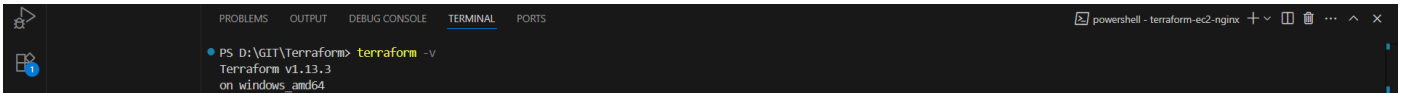
Terraform Task-2

Task Description:

Create 2 EC2 instances on 2 different regions and install nginx using terraform script.

Solution:

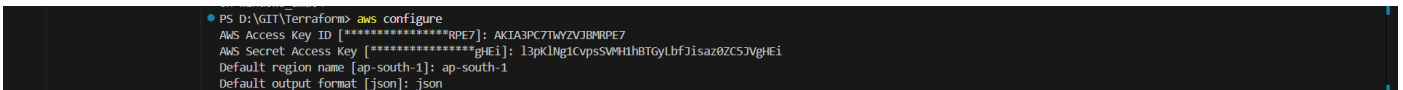
✓ 1. Install Terraform



```
PS D:\GIT\Terraform> terraform -v
Terraform v1.13.3
on windows_amd64
```

- Terraform is required to define and deploy AWS infrastructure.

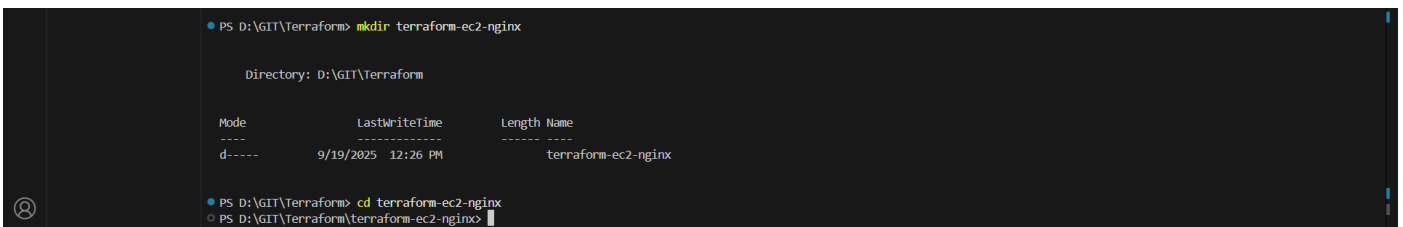
✓ 2. Install & Configure AWS CLI



```
PS D:\GIT\Terraform> aws configure
AWS Access Key ID [AKIA3PC7IMVZVJBMRPE7]: AKIA3PC7IMVZVJBMRPE7
AWS Secret Access Key [*****gHeI]: 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
Default region name [ap-south-1]: ap-south-1
Default output format [json]: json
```

- Terraform uses AWS credentials to access your account.

✓ 3. Create Terraform Project Directory



```
PS D:\GIT\Terraform> mkdir terraform-ec2-nginx

Directory: D:\GIT\Terraform

Mode                LastWriteTime         Length Name
----                -
d-----          9/19/2025 12:26 PM             terraform-ec2-nginx

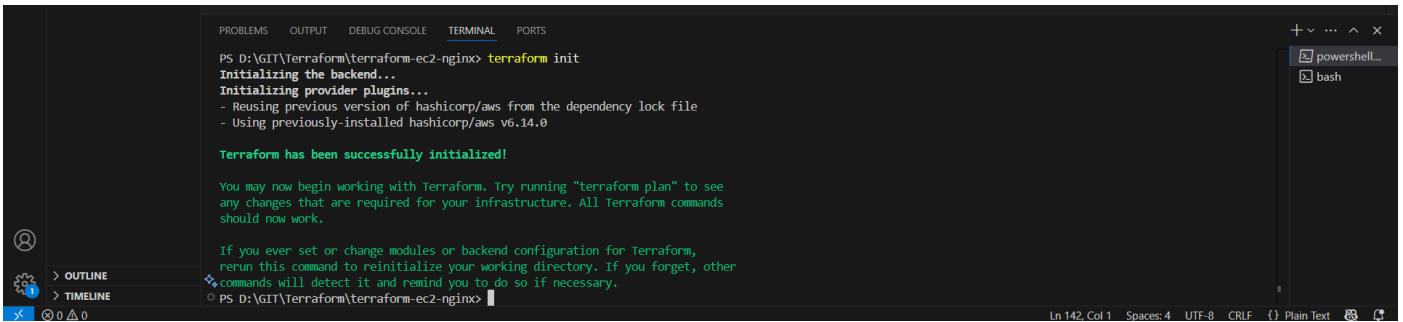
PS D:\GIT\Terraform> cd terraform-ec2-nginx
PS D:\GIT\Terraform\terraform-ec2-nginx>
```

- Organize all Terraform files.

✓ 4. Write main.tf

- This defines the **AWS providers, EC2 instances, security groups, key pairs, and Nginx installation.**
- I have attached my **main.tf** file in my Git repository.

✓ 5. Initialize Terraform



```
PS D:\GIT\Terraform\terraform-ec2-nginx> terraform init
Initializing the backend...
Initializing provider plugins...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Using previously-installed hashicorp/aws v6.14.0

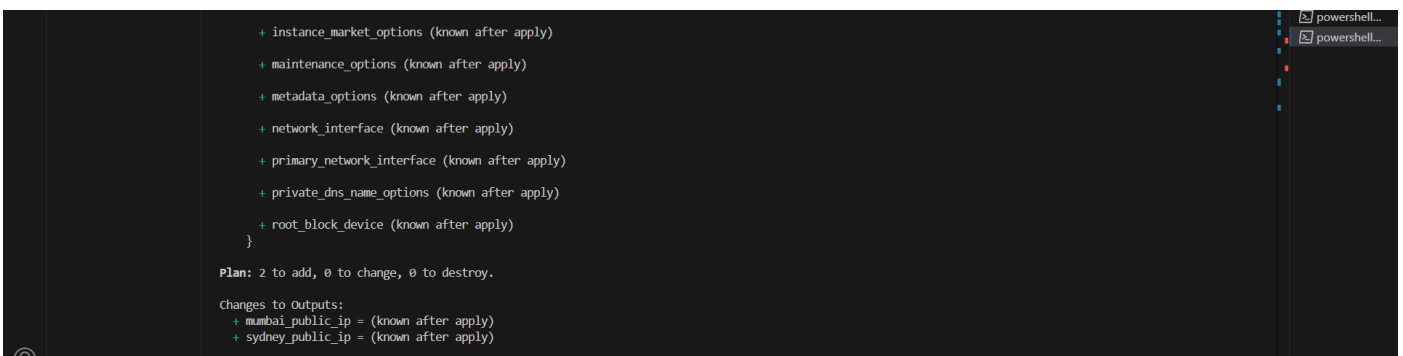
Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
PS D:\GIT\Terraform\terraform-ec2-nginx>
```

- Downloads provider plugins and prepares Terraform directory.

✓ 6. Review Terraform Plan



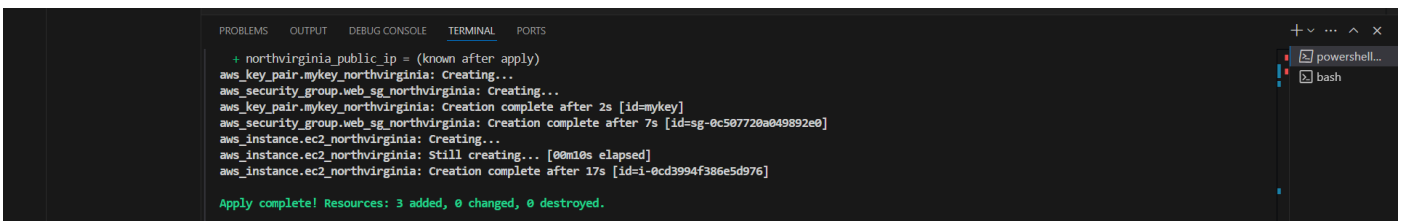
```
+ instance_market_options (known after apply)
+ maintenance_options (known after apply)
+ metadata_options (known after apply)
+ network_interface (known after apply)
+ primary_network_interface (known after apply)
+ private_dns_name_options (known after apply)
+ root_block_device (known after apply)
}

Plan: 2 to add, 0 to change, 0 to destroy.

Changes to Outputs:
+ mumbai_public_ip = (known after apply)
+ sydney_public_ip = (known after apply)
```

- See what resources will be created before deploying

✓ 7. Apply Terraform Plan



```
+ northvirginia_public_ip = (known after apply)
aws_key_pair.mykey_northvirginia: Creating...
aws_security_group.web_sg_northvirginia: Creating...
aws_key_pair.mykey_northvirginia: Creation complete after 2s [id=mykey]
aws_security_group.web_sg_northvirginia: Creation complete after 7s [id=sg-0c507720a049892e0]
aws_instance.ec2_northvirginia: Creating...
aws_instance.ec2_northvirginia: Still creating... [00m10s elapsed]
aws_instance.ec2_northvirginia: Creation complete after 17s [id=i-0cd3994f386e5d976]

Apply complete! Resources: 3 added, 0 changed, 0 destroyed.
```

- Deploy all AWS resources (EC2, SG, KeyPair).

✓ 8. Verify EC2 Instances

```
PS D:\GIT\Terraform\Terraform-ec2-nginx> terraform output
mumbai_public_ip = "65.1.95.179"
northvirginia_public_ip = "18.212.235.96"
PS D:\GIT\Terraform\Terraform-ec2-nginx>
```

The screenshot shows the AWS Management Console for the Mumbai region. The left sidebar lists navigation options like Dashboard, Events, Instances, Instance Types, Launch Templates, Spot Requests, Savings Plans, Reserved Instances, Dedicated Hosts, Capacity Reservations, Images, and Elastic Block Store. The main content area displays a table of EC2 instances. One instance, 'Mumbai-Nginx' with ID 'i-03f8a77ec648614f4', is highlighted as 'Running'. Below the table, the details for this instance are shown, including its public IPv4 address (65.1.95.179), private IPv4 address (172.31.5.139), and public DNS name (ec2-65-1-95-179.ap-south-1.compute.amazonaws.com).

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone
Mumbai-Instance	i-0aa6766e579ad43ec	Terminated	t2.micro	-	View alarms +	ap-south-1b
Mumbai-Instance	i-01268cc9b73ef5067	Terminated	t2.micro	-	View alarms +	ap-south-1b
Mumbai-Nginx	i-03f8a77ec648614f4	Running	t2.micro	2/2 checks passed	View alarms +	ap-south-1b

i-03f8a77ec648614f4 (Mumbai-Nginx)

Instance summary

Instance ID i-03f8a77ec648614f4	Public IPv4 address 65.1.95.179 open address	Private IPv4 addresses 172.31.5.139
IPv6 address -	Instance state Running	Public DNS ec2-65-1-95-179.ap-south-1.compute.amazonaws.com open address

The screenshot shows the AWS Management Console for the North Virginia region. The left sidebar lists navigation options like Dashboard, EC2 Global View, Events, Instances, Instance Types, Launch Templates, Spot Requests, Savings Plans, Reserved Instances, Dedicated Hosts, Capacity Reservations, Images, and Elastic Block Store. The main content area displays a table of EC2 instances. Two instances are listed: 'east-ubuntu' and 'NorthVirginia-Nginx'. The 'NorthVirginia-Nginx' instance with ID 'i-0cd3994f386e5d976' is highlighted as 'Running'. Below the table, the details for this instance are shown, including its public IPv4 address (18.212.235.96), private IPv4 address (172.31.27.230), and public DNS name (ec2-18-212-235-96.us-east-1.compute.amazonaws.com).

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IP
east-ubuntu	i-06cde6c22f9b3cb77	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1a	ec2-34-201-100-100.us-east-1.compute.amazonaws.com
NorthVirginia-Nginx	i-0cd3994f386e5d976	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1a	ec2-18-212-235-96.us-east-1.compute.amazonaws.com

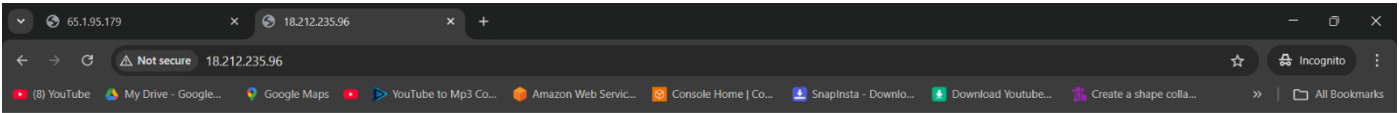
i-0cd3994f386e5d976 (NorthVirginia-Nginx)

Instance summary

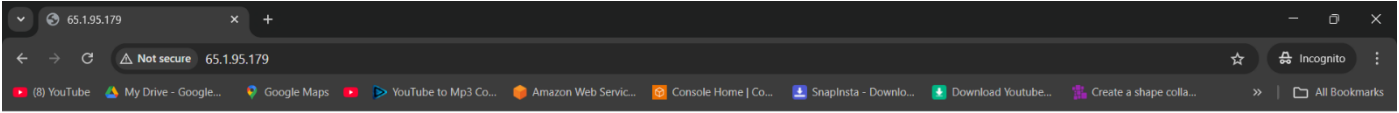
Instance ID i-0cd3994f386e5d976	Public IPv4 address 18.212.235.96 open address	Private IPv4 addresses 172.31.27.230
IPv6 address -	Instance state Running	Public DNS ec2-18-212-235-96.us-east-1.compute.amazonaws.com open address

- Check in AWS Console: Two EC2 instances are created (Mumbai & North Virginia).
- Get Public IP via Terraform Output

✓ 9. Test Nginx



Hello from Nginx - North Virginia



Hello from Nginx - Mumbai

- Ensure Nginx is running and serving page.
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